fn approximate_to_tradeoff

Aishwarya Ramasethu (@aramasethu), Yu-Ju Ku (@yuju1998), Jordan Awan, Michael Shoemate (@Shoeboxam)

April 29, 2026

This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `epsilon` of type `f64`
- Argument `delta` of type `f64`

Caller-verified

None

Pseudocode

```python
def appproximate_to_tradeoff(
    param: tuple[f64, f64]
) -> tuple[Callable[[RBig], RBig], RBig]:
    epsilon, delta = param

    exp_eps = epsilon.with_rounding(Down).exp()  #
    exp_eps = RBig.try_from(exp_eps)

    exp_neg_eps = (-epsilon).with_rounding(Up).exp()  #
    exp_neg_eps = RBig.try_from(exp_neg_eps)

    fixed_point = (RBig(1) - delta) / (RBig(1) + exp_eps)

    if fixed_point >= RBig(1, 2):
        raise ValueError("fixed point of tradeoff curve must be less than 1/2")

    def tradeoff(alpha: RBig) -> RBig:  #
        t1 = RBig(1) - delta - exp_eps * alpha
        t2 = exp_neg_eps * (RBig(1) - delta - alpha)
        return max(max(t1, t2), RBig(0))

    return tradeoff, fixed_point
```

Postcondition

Theorem 1.1. Given a pair of epsilon and delta, the pseudocode returns an error if epsilon or delta are invalid, otherwise returns the corresponding symmetric nontrivial f-DP tradeoff curve with conservative arithmetic.

Proof. We start with the following alternative definition of DP from Dong, Roth, and Su 2022.

Definition 1.2 (Dong, Roth, and Su 2022, Proposition 2.5)**.** Let $\epsilon > 0$ and $\delta \geq 0$, and define

$$f_{\epsilon,\delta}(\alpha) = \max(0, 1 - \delta - \exp(\epsilon)\alpha, \exp(-\epsilon)(1 - \delta - \alpha)). \tag{1}$$

Then we say that a mechanism M satisfies (ϵ, δ)-DP if it satisfies $f_{\epsilon,\delta}$-DP.

The definition assumes $\alpha \in [0, 1]$.

On lines 6 and 9, arithmetic is computed in a manner which results in over-estimates of the privacy loss, and therefore a tradeoff curve that bows further away from $1-\alpha$. The results of these constants are converted exactly into rationals to be used in the tradeoff function. The function defined on line 17 implements the formula in Definition 1.2 with exact fractional arithmetic. The implementation of the tradeoff function in the pseudocode is the corresponding symmetric nontrivial f-DP tradeoff curve with conservative arithmetic, satisfying the postcondition. □

References

Dong, Jinshuo, Aaron Roth, and Weijie J. Su (2022). "Gaussian Differential Privacy". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84.1, pp. 3–37. ISSN: 1369-7412.